

Edeon — Registration Readiness Dossier

Compound: Glyphosate (CAS: 1071-83-6)

SMILES: O=C(O)CNC(=O)O

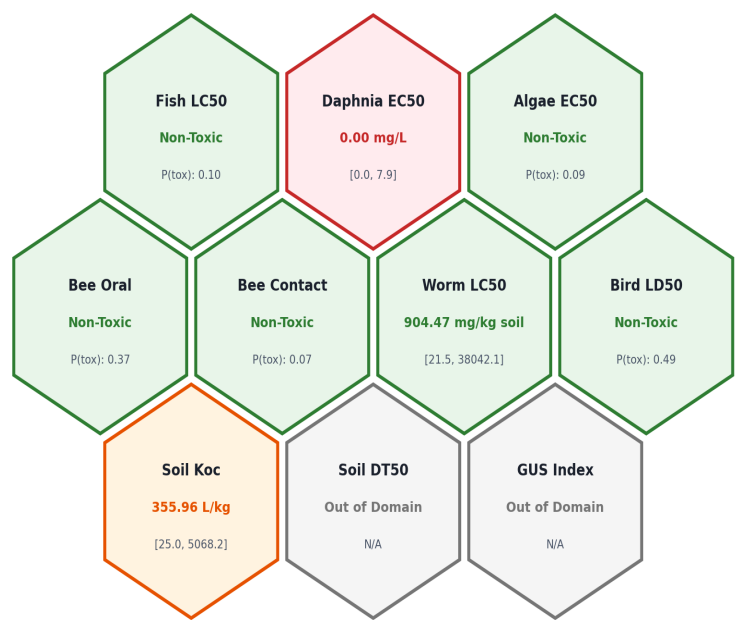
REGISTRATION READINESS VERDICT: NO_GO

Driver: Showstopper triggered: Aquatic hazard (CLP H400/H410/H411/H412) | **Confidence:** moderate

Rationale: Most sensitive endpoint: Daphnia EC50 = 0.0003 mg/L, Not rapidly degradable (default assumption)

Tier-1 Predictive Honeycomb

Glyphosate — QSAR Safety Profile



Regulatory Cut-Offs & Criteria Evaluation

Criterion / Endpoint	Status	Evidence & Details
Genotoxicity / Mutagenicity alerts (Benigni-Bossa)	PASS	No genotoxicity structural alerts detected
Endocrine disruption screening (structural)	PASS	No endocrine disruption structural alerts detected
Skin sensitization alerts (protein reactivity)	PASS	No skin sensitization alerts detected
PBT/vPvB (REACH Annex XIII)	WATCH	Daphnia EC50 = 0.0003 mg/L < 0.1 (T)
Groundwater concern (EU DWD 0.1 µg/L trigger)	PASS	GUS = 1.70 < 1.8 (non-leacher); GUS classification: non-leacher
Aquatic hazard (CLP H400/H410/H411/H412)	LIKELY_SHO WSTOPPER	Most sensitive endpoint: Daphnia EC50 = 0.0003 mg/L; Not rapidly degradable (default assumption)
Acute mammalian toxicity (CLP oral)	PASS	Predicted rat oral LD50 = 218776 mg/kg
Soil persistence (EU 1107/2009)	PASS	Predicted soil DT50 = 15 days; DT50 ≤ 90 d — below persistence trigger

IN-SILICO SCREENING ONLY — These results are computational triage signals based on predicted endpoints and structural pattern matching. They are NOT regulatory determinations and cannot replace experimental studies or expert regulatory assessment.